	so pulses / energy <u>build up</u> the amplitude (don't accept maximum amplitude)	A1
3ai	Transverse	A1
3aii	$t = \dfrac{\text{distance}}{\text{speed}}$ $= \dfrac{4.4 \times 10^{12}}{3.0 \times 10^{8}}$ $= 1.5 \times 10^{4}$ s	A1
3aiii	power received by dish aerial = intensity received × area of dish $P = \dfrac{\text{output (source) power}}{4\pi r^2} \times 280$ $= \dfrac{25}{4\pi(4.4 \times 10^{12})^2} \times 280$ $= 2.9 \times 10^{-23}$ W	C1 C1 A1
3aiv	The actual power is greater because the <u>transmitting aerial is directing power towards the Earth</u> instead of radiating uniformly in all directions as assumed in **(a)(iii)**.	A1
3bi	$v = f\lambda$ $\lambda = 0.040 \times 1.5$ $\lambda = 0.060$ m	C1 A1
3bii	path difference = (44 – 29) / 6.0 = 2.5 λ **either** waves have path difference = (n + ½) λ **or** waves have phase difference = π so destructive interference	M1 M1 A1
3ci	Incident sound wave from loudspeaker travels to the water surface and is reflected. The incident and reflected waves superpose / interfere / overlap to form a stationary wave.	B1 B1
3cii	distance = λ / 2 = 0.090 m	A1
3ciii	letter X shown at level B and at level A	A1
4a	Electric field lines are radial/normal to surface (of sphere)	B1